

NCT Number: NCT02477826 (CheckMate-227)

Eligibility Criteria:

- Subjects with histologically confirmed Stage IV or recurrent NSCLC squamous or non-squamous histology, with no prior systemic anticancer therapy
- Subjects must have programmed death-ligand 1 (PD -L1) immunohistochemical (IHC) testing, with results, performed by the central lab during the Screening period
- Eastern Cooperative Oncology Group (ECOG) Performance Status of ≤ 1
- Measurable disease by CT or MRI per response evaluation criteria in solid tumors version 1.1 (RECIST 1.1) criteria

Ineligibility Criteria:

- Subjects with untreated Central nervous system (CNS) metastases are excluded
- Subjects with an active, known or suspected autoimmune disease are excluded
- Any positive test for hepatitis B virus or hepatitis C virus or human immunodeficiency virus (HIV) indicating acute or chronic infection

NCT Number: NCT02613507 (CheckMate-078)

Eligibility Criteria:

- Disease progression experienced during or after one prior platinum containing doublet chemotherapy
- Stage IIIb/IV or recurrent disease
- Male and Female ≥ 18 years of age
- Measurable disease per RECIST 1.1
- Performance Status ≤ 1

Ineligibility Criteria:

- History of Carcinomatous meningitis
- Active Central nervous system (CNS) metastases
- History of auto immune diseases
- Prior treatment with Docetaxel
- Prior treatment with ipilimumab or any drug targeting T-Cell costimulation or checkpoint pathways

NCT Number: NCT02220894 (KEYNOTE-042)

Eligibility Criteria:

- Histologically- or cytologically-confirmed diagnosis of advanced or metastatic NSCLC
- PD-L1 positive tumor
- Measureable disease based on Response Evaluation Criteria in Solid Tumors (RECIST) 1.1
- Life expectancy of at least 3 months
- No prior systemic chemotherapy for the treatment of the participant's advanced or metastatic disease (treatment with chemotherapy and/or radiation as part of neoadjuvant/adjuvant therapy is allowed as long as completed at least 6 months prior to diagnosis of advanced or metastatic disease)
- Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1
- Adequate organ function
- No prior malignancy, with the exception of basal cell carcinoma of the skin, superficial bladder cancer, squamous cell carcinoma of the skin, or in situ cancer, or has undergone potentially curative therapy with no evidence of that disease recurrence for 5 years since initiation of that therapy
- Submission of formalin-fixed diagnostic tumor tissue (in the case of participants having received adjuvant systemic therapy, the tissue should be taken after completion of this therapy)
- Female participants of childbearing potential must have a negative urine or serum pregnancy test and must be willing to use two adequate barrier methods of contraception or a barrier method plus a hormonal method starting with the screening visit through 120 days after the last dose of pembrolizumab or 180 days after the last dose of chemotherapeutic agents used in the study
- Male participants with a female partner(s) of child-bearing potential must be willing to use two adequate barrier methods of contraception from screening through 120 days after the last dose of pembrolizumab or 180 days after the last dose of chemotherapeutic agents used in the study

Ineligibility Criteria:

- Epidermal growth factor receptor (EGFR)-sensitizing mutation and/or is echinoderm microtubule-associated protein-like 4(EML4) gene/anaplastic lymphoma kinase (ALK) gene fusion positive
- Currently participating or has participated in a study of an investigational agent or using an investigational device within 4 weeks of the first dose of study therapy
- No tumor specimen evaluable for PD-L1 expression by the central study laboratory
- Squamous histology and received carboplatin in combination with paclitaxel in the adjuvant setting
- Is receiving systemic steroid therapy ≤ 3 days prior to the first dose of study therapy or receiving any other form of immunosuppressive medication with the exception of daily steroid replacement therapy
- The NSCLC can be treated with curative intent with either surgical resection and/or chemoradiation

- Expected to require any other form of systemic or localized antineoplastic therapy while on study
- Any prior systemic cytotoxic chemotherapy, biological therapy or major surgery within 3 weeks of the first dose of study therapy; received lung radiation therapy >30 Gy within 6 months of the first dose of study therapy
- Prior therapy with an anti-PD-1, anti-PD-L1, anti-PD-L2, anti-CD137, or anti-cytotoxic T-lymphocyte-associated antigen-4 (CTLA-4) antibody (including ipilimumab or any other antibody or drug specifically targeting T-cell co-stimulation or checkpoint pathways)
- Known central nervous system metastases and/or carcinomatous meningitis
- Active autoimmune disease that has required systemic treatment in the past 2 years
- Had allogeneic tissue/solid organ transplantation
- Interstitial lung disease or history of pneumonitis that has required oral or IV steroids
- Has received or will receive a live vaccine within 30 days prior to the first study therapy (seasonal flu vaccines that do not contain live vaccine are permitted)
- Active infection requiring intravenous systemic therapy
- Known history of human immunodeficiency virus (HIV)
- Known active Hepatitis B or C
- Regular user (including "recreational use") of any illicit drugs or had a recent history (within the last year) of substance abuse (including alcohol)
- Pregnant, breastfeeding, or expecting to conceive or father children within the projected duration of the study

NCT Number: NCT02775435 (KEYNOTE-407)

Eligibility Criteria:

- Has a histologically or cytologically confirmed diagnosis of stage IV (M1a or M1b-American Joint Committee on Cancer [AJCC] 7th edition) squamous NSCLC.
- Has measurable disease based on RECIST 1.1 as determined by the local site investigator/radiology assessment.
- Has not received prior systemic treatment for metastatic NSCLC.
- Has provided tumor tissue from locations not radiated prior to biopsy.
- Has a life expectancy of at least 3 months.
- Has a performance status of 0 or 1 on the Eastern Cooperative Oncology Group (ECOG) Performance Status.
- Has adequate organ function.
- If female of childbearing potential, is willing to use an adequate method of contraception for the course of the study through 180 days after the last dose of study drug.
- If male with a female partner(s) of child-bearing potential, must agree to use an adequate method of contraception starting with the first dose of study drug through 95 days after the last dose of study drug. Males with pregnant partners must agree to use a condom; no additional method of contraception is required for the pregnant partner.

Ineligibility Criteria:

- Has non-squamous histology NSCLC.
- Is currently participating and receiving study therapy or has participated in a study of an investigational agent and received study therapy or used an investigational device within 4 weeks prior to administration of pembrolizumab.
- Before the first dose of study drug: a) Has received prior systemic cytotoxic chemotherapy for metastatic disease; b) Has received other targeted or biological antineoplastic therapy (e.g., erlotinib, crizotinib, cetuximab) for metastatic disease; c) Has had major surgery (<3 weeks prior to first dose).
- Received radiation therapy to the lung that is > 30 Gy within 6 months of the first dose of study drug.
- Completed palliative radiotherapy within 7 days of the first dose of study drug.
- Is expected to require any other form of antineoplastic therapy while on study.
- Has received a live-virus vaccination within 30 days of planned treatment start.
- Has a known history of prior malignancy except if the participant has undergone potentially curative therapy with no evidence of that disease recurrence for 5 years since initiation of that therapy.
- Has known active central nervous system (CNS) metastases and/or carcinomatous meningitis.

- Has pre-existing peripheral neuropathy that is $\geq$ Grade 2 by Common Terminology Criteria for Adverse Events (CTCAE) version 4 criteria.
- Previously had a severe hypersensitivity reaction to treatment with another monoclonal antibody.
- Has a known sensitivity to any component of carboplatin or paclitaxel or nab-paclitaxel.
- Has active autoimmune disease that has required systemic treatment in past 2 years.
- Is on chronic systemic steroids.
- Had prior treatment with any other anti-programmed cell death 1 (anti-PD-1), or programmed cell death ligand 1 (PD-L1) or PD-L2 agent or an antibody or a small molecule targeting other immuno-regulatory receptors or mechanisms.
- Has participated in any other pembrolizumab trial and has been treated with pembrolizumab.
- Has an active infection requiring therapy.
- Has known history of Human Immunodeficiency Virus (HIV).
- Has known active Hepatitis B or C. Active Hepatitis B.
- Is, at the time of providing documented informed consent, a regular user (including "recreational use") of any illicit drugs or has a recent history (within the last year) of substance abuse (including alcohol).
- Has interstitial lung disease or a history of pneumonitis that required oral or intravenous glucocorticoids to assist with management.

NCT Number: NCT02366143 (IMpower150)

Eligibility Criteria:

- Eastern Cooperative Oncology Group performance status 0 or 1
- Histologically or cytologically confirmed, Stage IV non-squamous NSCLC
- Participants with no prior treatment for Stage IV non-squamous NSCLC
- Known PD-L1 status as determined by immunohistochemistry assay performed on previously obtained archival tumor tissue or tissue obtained from a biopsy at screening
- Measurable disease as defined by RECIST v1.1
- Adequate hematologic and end organ function

Ineligibility Criteria:

- Cancer-Specific Exclusions:
 - Active or untreated central nervous system metastases
 - Malignancies other than NSCLC within 5 years prior to randomization, with the exception of those with a negligible risk of metastasis or death treated with expected curative outcome
- General Medical Exclusions:
 - Pregnant or lactating women
 - History of autoimmune disease
 - History of idiopathic pulmonary fibrosis, organizing pneumonia, drug-induced pneumonitis, idiopathic pneumonitis, or evidence of active pneumonitis on screening chest computed tomography scan. History of radiation pneumonitis in the radiation field (fibrosis) is permitted
 - Positive test for human immunodeficiency virus
 - Active hepatitis B or hepatitis C
 - Severe infection within 4 weeks prior to randomization
 - Significant cardiovascular disease
 - Illness or condition that interferes with the participant's capacity to understand, follow and/or comply with study procedures
 - Prior treatment with cluster of differentiation 137 agonists or immune checkpoint blockade therapies, anti-programmed death-1, and anti-PD-L1 therapeutic antibodies

NCT Number: NCT02486718 (IMpower010)

Eligibility Criteria:

- Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1
- Histological or cytological diagnosis of Stage IB (tumors greater than or equal to $\geq$ 4 centimeters [cm])-IIIA (T2-3 N0, T1-3 N1, T1-3 N2, T4 N0-1) NSCLC (per the Union Internationale Contre le Cancer staging system (UICC)/American Joint Committee on Cancer staging system (AJCC) staging system, 7th edition; Detterbeck et al. 2009)
- Participants must have had complete resection of NSCLC 4-12 weeks ($\geq$ 28 days and less than or equal to $\leq$ 84 days) prior to enrollment and must be adequately recovered from surgery
- If mediastinoscopy was not performed preoperatively, it is required that, at a minimum, mediastinal lymph node systematic sampling will have occurred. Systematic sampling is defined as removal of at least one representative lymph node at specified levels. MLND entails resection of all lymph nodes at those same levels. For a right thoracotomy, sampling or MLND is required at levels 4 and 7 and for a left thoracotomy, levels 5 and/or 6 and 7. Exceptions will be granted if there is clear documentation in the operative report or in a separately submitted addendum by the surgeon of exploration of the required lymph node areas, the participant will be considered eligible if no lymph nodes are found in those areas; if participants have documented N2 disease in one level (per the UICC/AJCC staging system, 7th edition; Detterbeck et al. 2009), not all levels need to be sampled; if the preoperative staging imaging results (contrast computed tomography [CT] and positron emission tomography [PET] scans) do not suggest evidence of disease in the mediastinum, the participant will be considered eligible if N2 nodal sampling is not performed per surgeon's decision
- Eligible to receive a cisplatin-based chemotherapy regimen
- Adequate hematologic and end-organ function
- Women who are not postmenopausal ($\geq$ 12 months of non-therapy-induced amenorrhea) or surgically sterile must have a negative serum pregnancy test result within 14 days prior to initiation of cisplatin-based chemotherapy

Ineligibility Criteria:

- Illness or condition that may interfere with a participant's capacity to understand, follow, and/or comply with study procedures
- Pregnant and lactating women
- Treatment with prior systemic chemotherapy: Chemotherapy for early stage of malignancy with curative intent, provided that the last dose received was more than 5 years prior to enrollment and low-dose chemotherapy for non-malignant conditions may be allowed upon approval by the Medical Monitor

- Hormonal cancer therapy or radiation therapy as prior cancer treatment within 5 years before enrollment
- Treatment with any other investigational agent with therapeutic intent within 28 days prior to enrollment
- Participants with hearing impairment
- Known sensitivity to any component of the chemotherapy regimen the participant will be assigned to, or to mannitol
- Prior treatment with cluster of differentiation (CD) 137 (CD137) agonists or immune checkpoint blockade therapies, anti-programmed death-1 (PD-1), and anti programmed death ligand 1 (PD-L1) therapeutic antibodies
- Malignancies other than NSCLC within 5 years prior to randomization, with the exception of those with a negligible risk of metastasis or death (e.g., expected 5-year OS greater than [$>$] 90 percent [%]) treated with expected curative outcome (such as adequately treated carcinoma in situ of the cervix, basal or squamous cell skin cancer, localized prostate cancer treated surgically with curative intent, ductal carcinoma in situ treated surgically with curative intent)
- History of severe allergic, anaphylactic, or other hypersensitivity reactions to chimeric or humanized antibodies or fusion proteins
- Known hypersensitivity to biopharmaceuticals produced in Chinese hamster ovary cells or any component of the atezolizumab formulation
- History of autoimmune disease, including but not limited to myasthenia gravis, myositis, autoimmune hepatitis, systemic lupus erythematosus, rheumatoid arthritis, inflammatory bowel disease, vascular thrombosis associated with antiphospholipid syndrome, Wegener's granulomatosis, Sjögren's syndrome, Guillain-Barré syndrome, multiple sclerosis, vasculitis, or glomerulonephritis
- Positive test for human immunodeficiency virus (HIV)
- Participants with active hepatitis B (chronic or acute; defined as having a positive hepatitis B surface antigen [HBsAg] test at screening) or hepatitis C
- Active tuberculosis
- Significant cardiovascular disease, such as New York Heart Association cardiac disease (Class II or greater), myocardial infarction, or cerebrovascular accident within the previous 3 months, unstable arrhythmias, or unstable angina
- History of idiopathic pulmonary fibrosis, organizing pneumonia (e.g., bronchiolitis obliterans), drug-induced pneumonitis, idiopathic pneumonitis, or evidence of active pneumonitis on screening chest CT scan
- Prior allogeneic bone marrow transplantation or solid organ transplant
- Any other diseases, metabolic dysfunction, physical examination finding, or clinical laboratory finding giving reasonable suspicion of a disease or condition that contraindicates the use of an investigational drug or that

- may affect the interpretation of the results or render the participant at high risk from treatment complications
- Known tumor PD-L1 expression status as determined by an immunohistochemistry (IHC) assay from other clinical studies (e.g., participants whose PD-L1 expression status was determined during screening for entry into a study with anti-PD-1 or anti-PD-L1 antibodies but were not eligible are excluded)
 - Specific Exclusions for Pemetrexed Treatment
 - - Participants with squamous cell histology
 - Signs or symptoms of infection within 14 days prior to randomization (severe infection within 28 days prior to randomization), including but not limited to hospitalization for complications of infection, bacteremia, or severe pneumonia
 - Received therapeutic oral or intravenous (IV) antibiotics within 14 days prior to randomization
 - Major surgical procedure within 28 days prior to randomization or anticipation of need for a major surgical procedure during the course of the study
 - Administration of a live, attenuated vaccine within 4 weeks prior to initiation of study treatment or anticipation that such a live attenuated vaccine will be required during the study
 - Treatment with systemic immunostimulatory agents (including but not limited to interferons or interleukin-2) within 4 weeks or 5 half-lives of the drug, whichever is longer, prior to randomization: Prior treatment with cancer vaccines is allowed
 - Treatment with systemic corticosteroids or other immunosuppressive medications (including but not limited to prednisone, dexamethasone, cyclophosphamide, azathioprine, methotrexate, thalidomide, and anti-tumor necrosis factor [anti-TNF] agents) within 14 days prior to randomization

NCT Number: NCT02075840 (ALEX)

Eligibility Criteria:

- Histologically or cytologically confirmed diagnosis of advanced or recurrent (Stage IIIB not amenable for multimodality treatment) or metastatic (Stage IV) NSCLC that is ALK-positive as assessed by the Ventana immunohistochemistry (IHC) test
- Life expectancy of at least 12 weeks
- Eastern cooperative oncology group performance status (ECOG PS) of 0-2
- Participants with no prior systemic treatment for advanced or recurrent (Stage IIIB not amenable for multimodality treatment) or metastatic (Stage IV) NSCLC
- Adequate renal, and hematologic function
- Participants must have recovered from effects of any major surgery or significant traumatic injury at least 28 days before the first dose of study treatment
- Measurable disease by response evaluation criteria in solid tumors (RECIST) version 1.1 (v1.1) prior to the administration of study treatment
- Prior brain or leptomeningeal metastases allowed if asymptomatic (e.g., diagnosed incidentally at study baseline)
- Negative pregnancy test for all females of child bearing potential
- Use of highly effective contraception as defined by the study protocol

Ineligibility Criteria:

- Participants with a previous malignancy within the past 3 years
- Any gastrointestinal (GI) disorder or liver disease
- National cancer institute common terminology criteria for adverse events (NCI CTCAE) (version 4.0) Grade 3 or higher toxicities due to any prior therapy (e.g., radiotherapy) (excluding alopecia)
- History of organ transplant
- Co-administration of anti-cancer therapies other than those administered in this study
- Participants with baseline QTc greater than (>) 470 milliseconds or symptomatic bradycardia
- Recipient of strong/potent cytochrome P4503A inhibitors or inducers within 14 days prior to the first dose until the end of study treatment
- Recipient of any drug with potential QT interval prolonging effects within 14 days prior to the first dose for all participants and while on treatment through the end of the study for crizotinib-treated participants only
- History of hypersensitivity to any of the additives in the alectinib and crizotinib drug formulation
- Pregnancy or lactation
- Any clinically significant disease or condition (or history of) that could interfere with, or for which the

treatment might interfere with, the conduct of the study or the absorption of oral medications or that would, in the opinion of the principal investigator, pose an unacceptable risk to the participant in this study

- Any psychological, familial, sociological, or geographical condition potentially hampering compliance with the study protocol requirements and/or follow-up procedures; those conditions should be discussed with the participant before trial entry

NCT Number: NCT02296125 (FLAURA)

Eligibility Criteria:

- Male or female, aged at least 18 years.
- Pathologically confirmed adenocarcinoma of the lung.
- Locally advanced or metastatic NSCLC, not amenable to curative surgery or radiotherapy.
- The tumour harbours one of the 2 common EGFR mutations known to be associated with EGFR-TKI sensitivity (Ex19del, L858R).
- Mandatory provision of an unstained, archived tumour tissue sample in a quantity sufficient to allow for central analysis of EGFR mutation status.
- Patients must be treatment-naïve for locally advanced or metastatic NSCLC and eligible to receive first-line treatment with gefitinib or erlotinib as selected by the participating centre. Prior adjuvant and neo-adjuvant therapy is permitted (chemotherapy, radiotherapy, investigational agents).
- Provision of informed consent prior to any study specific procedures, sampling, and analysis.
- World Health Organization Performance Status of 0 to 1 with no clinically significant deterioration over the previous 2 weeks and a minimum life expectancy of 12 weeks

Ineligibility Criteria:

- Treatment with any of the following:
- Prior treatment with any systemic anti-cancer therapy for locally advanced/metastatic NSCLC.
- Prior treatment with an EGFR-TKI.
- Major surgery within 4 weeks of the first dose of study drug.
- Radiotherapy treatment to more than 30% of the bone marrow or with a wide field of radiation within 4 weeks of the first dose of study drug.
- Patients currently receiving medications or herbal supplements known to be potent inducers of cytochrome P450 (CYP) 3A4.
- Alternative anti-cancer treatment
- Treatment with an investigational drug within five half-lives of the compound or any of its related material.
- Any concurrent and/or other active malignancy that has required treatment within 2 years of first dose of study drug.
- Spinal cord compression, symptomatic and unstable brain metastases, except for those patients who have completed definitive therapy, are not on steroids, have a stable neurologic status for at least 2 weeks after completion of the definitive therapy and steroids.
- Any evidence of severe or uncontrolled systemic diseases, including uncontrolled hypertension and active bleeding diatheses; or active infection including hepatitis B, hepatitis C and human immunodeficiency virus

(HIV).

- Refractory nausea and vomiting, chronic gastrointestinal diseases, inability to swallow the formulated product, or previous significant bowel resection that would preclude adequate absorption of AZD9291.
- Any of the following cardiac criteria:
 - Mean resting corrected QT interval (QTc) >470 msec, obtained from 3 ECGs, using the screening clinic ECG machine-derived QTcF value.
 - Any clinically important abnormalities in rhythm, conduction, or morphology of resting ECG.
 - Any patient with any factors that increase the risk of QTc prolongation or risk of arrhythmic events or unexplained sudden death under 40 years of age in first-degree relatives or any concomitant medication known to prolong the QT interval.
 - Past medical history of ILD, drug-induced ILD, radiation pneumonitis which required steroid treatment, or any evidence of clinically active ILD.
- Involvement in the planning and/or conduct of the study

NCT Number: NCT01774721 (ARCHER1050)

Eligibility Criteria:

- Evidence of histo or cytopathology confirmed, advanced NSCLC (with known histology) with the presence of EGFR-activating mutation (exon 19 deletion or the L858R mutation in exon 21).
- It is acceptable for subjects with the presence of the exon 20 T790M mutation together with either EGFR-activating mutation (exon 19 deletion or the L858R mutation in exon 21) to be included in this study
- No prior treatment with systemic therapy for locally advanced or metastatic NSCLC. Minimum of 12 months disease free interval between completion of neoadjuvant/adjuvant systemic therapy and recurrence of NSCLC
- Adequate tissue sample must be available for central analyses.
- Adequate renal, hematologic, liver function.
- ECOG PS of 0-1.
- Radiologically measurable disease.

Ineligibility Criteria:

- Any evidence of mixed histology that includes elements of small cell or carcinoid lung cancer.
- Any other mutation other than exon 19 deletion or L858R in exon 21, with or without the presence of the exon 20 T790M mutation.
- Any history of brain metastases or leptomeningeal metastases.
- Any previous anti-cancer systemic treatment of early, locally advanced, or metastatic NSCLC.
- Any surgery(not including minor procedures such as lymph node biopsy), palliative radiotherapy or pleurodesis within 2 weeks of baseline assessments
- Any clinically significant gastrointestinal abnormalities that may impair intake, transit or absorption of the study drug.
- Current enrollment in another therapeutic clinical study.
- History of, or currently suspected, diffuse non-infectious pneumonitis or interstitial lung disease
- Uncontrolled medical disorders.
- Prior malignancy and concurrent malignancy except for non melanoma skin cancer or in-situ cervical cancer with no evidence of active disease.
- Use of narrow therapeutic index drugs that are CYP2D6 substrates from screening to randomization.

NCT Number: NCT03043872 (CASPIAN)

Eligibility Criteria:

- Histologically or cytologically documented extensive disease. Brain metastases; must be asymptomatic or treated and stable off steroids and anti-convulsants for at least 1 month prior to study treatment.
- Suitable to receive a platinum-based chemotherapy regimen as 1st line treatment.
- Life expectancy ≥ 12 weeks at Day 1.
- ECOG 0 or 1 at enrolment.
- No prior exposure to immune-mediated therapy excluding therapeutic anticancer vaccines.

Ineligibility Criteria:

- Any history of radiotherapy to the chest prior to systemic therapy or planned consolidation chest radiation therapy (except palliative care outside of the chest).
- Paraneoplastic syndrome of autoimmune nature, requiring systemic treatment or clinical symptomatology suggesting worsening of PNS
- Active infection including tuberculosis, HIV, hepatitis B and C
- Active or prior documented autoimmune or inflammatory disorders
- Uncontrolled intercurrent illness, including but not limited to interstitial lung disease.